Building Materials

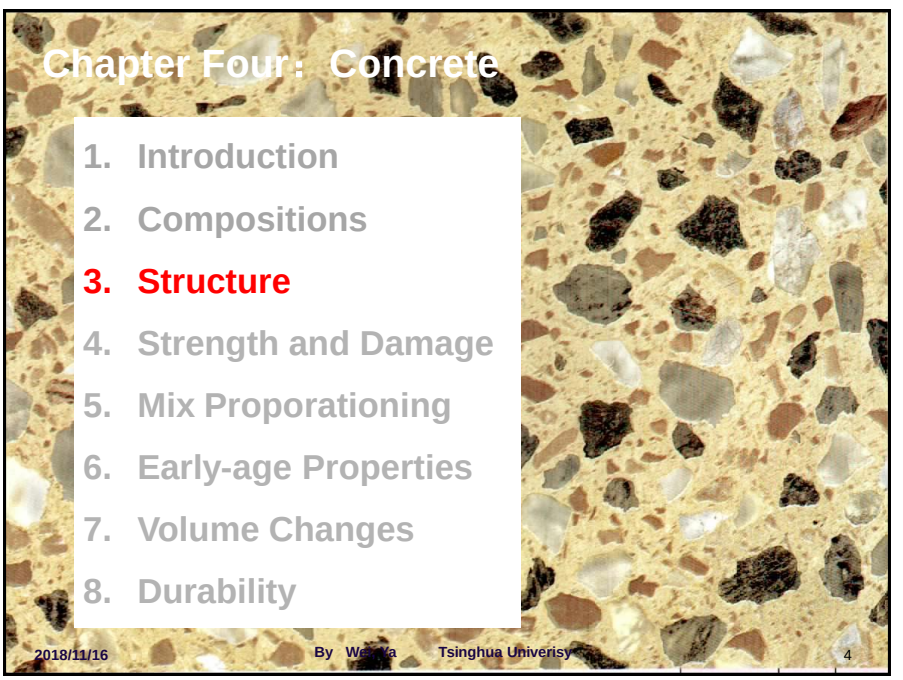
Wei, Ya Ph.D.
Department of Civil Engineering

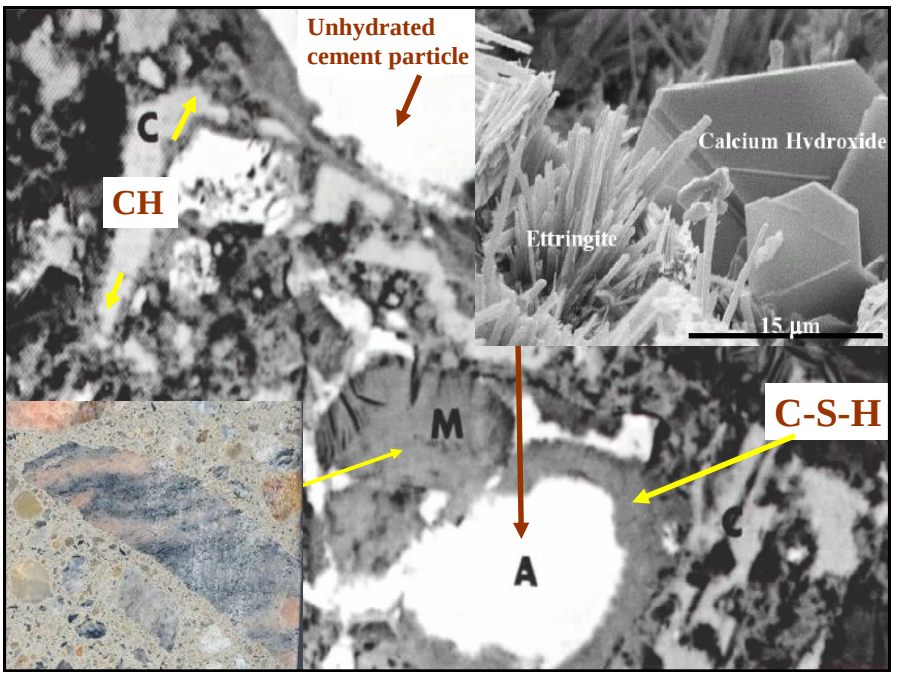
Tel. : +86-10-62771646
Email: yawei@tsinghua.edu.cn

Chapter One Introduction
Chapter Two The Fundamentals
Chapter Three Metal
Chapter Four Concrete
Chapter Five Asphalt
Chapter Six Masonry
Chapter Seven Wood

Chapter Four：Concrete

1. Introduction
2. **Compositions**
3. Structure
4. Strength and Damage
5. Mix Proporationing
6. Early-age Properties
7. Volume Changes
8. Durability



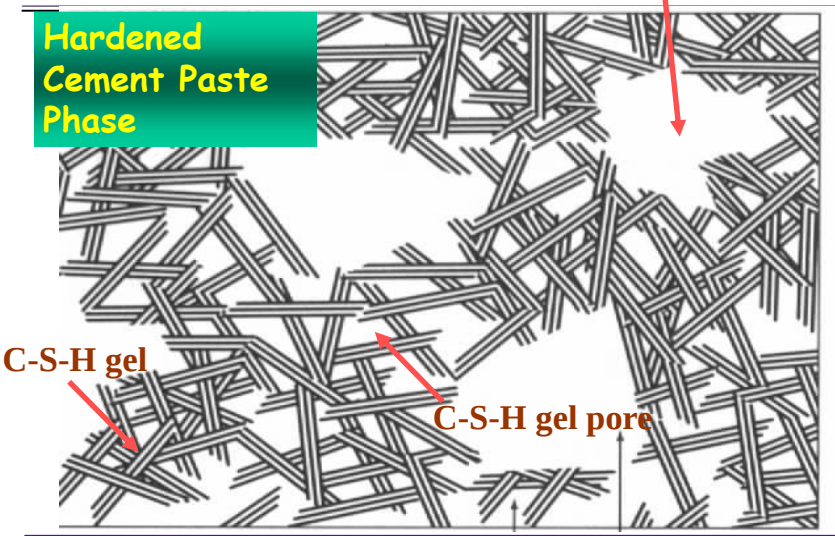


4.3 Structure

- Microstructure of Concrete is Complex and Heterogeneous
- Concrete has three phases:
  - Aggregate phase
  - Hardened cement paste phase
  - Transition zone phase
- Each of the three phases is itself multiphase in nature.
- Unlike other engineering materials, the structure of concrete does not remain stable.

4.3 Structure

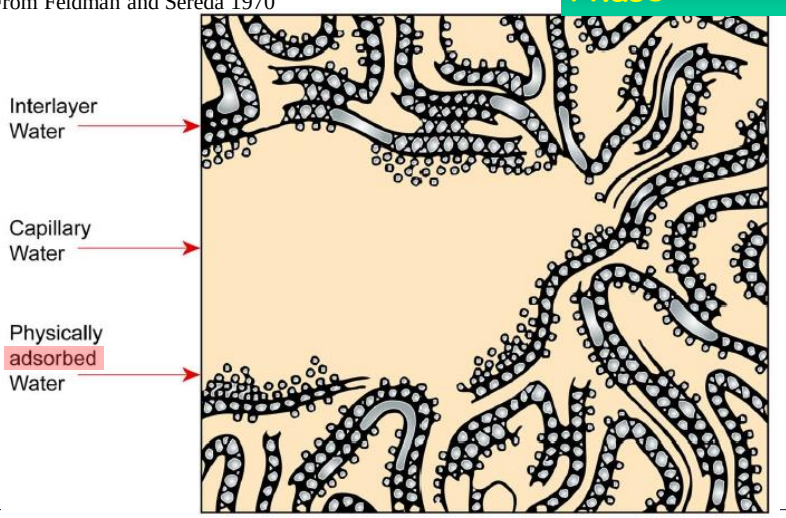
Hardened Cement Paste Phase

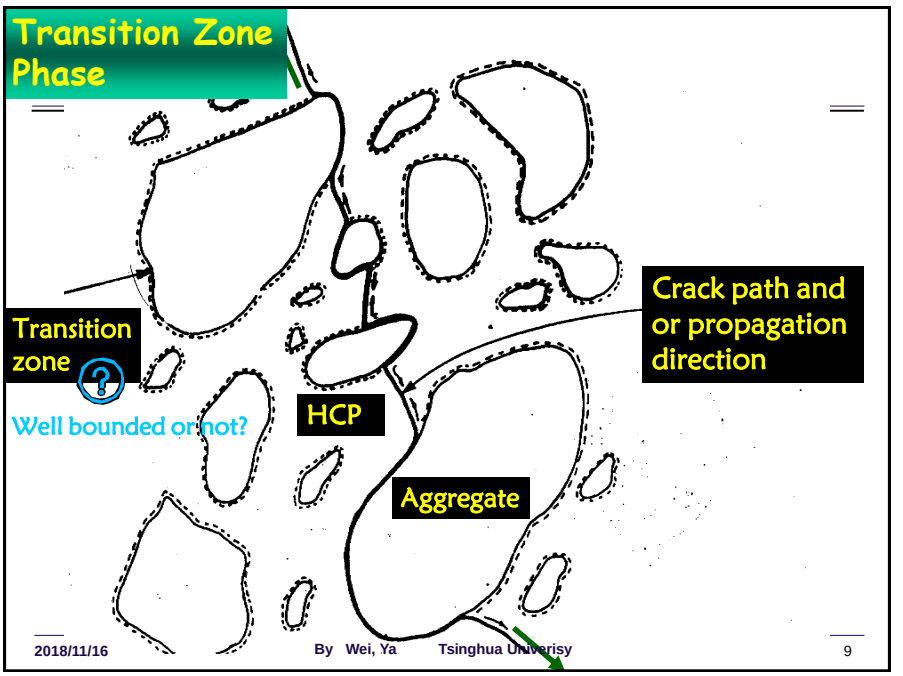


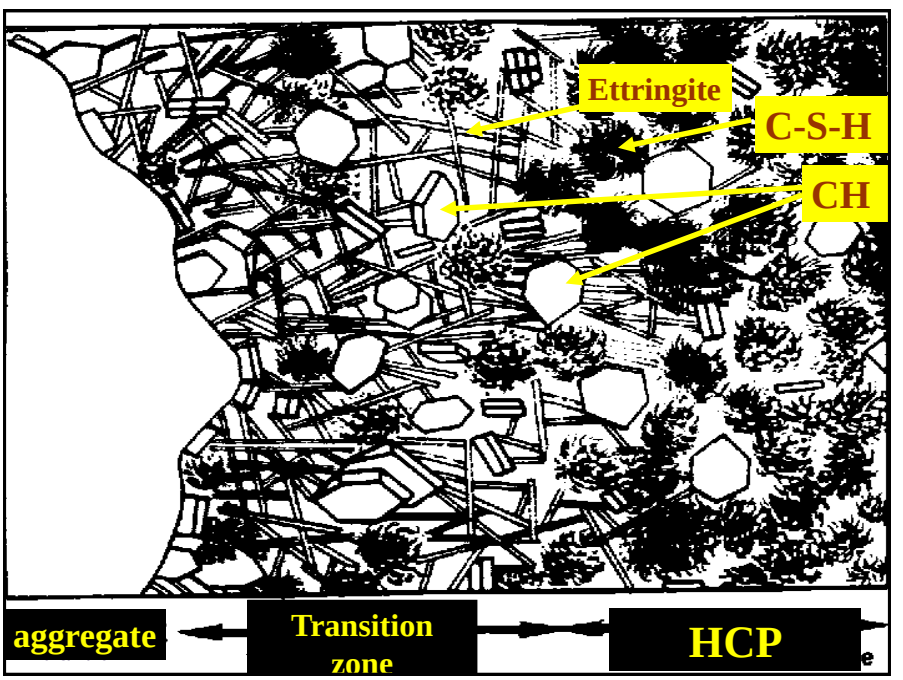
4.3 Structure

Hardened Cement Paste Phase

From Feldman and Sereda 1970




Transition Zone Phase
Transition zone
Well bounded or not?
HCP
Aggregate
Crack path and or propagation direction
2018/11/16
By Wei, Ya Tsinghua Universisy
9


Ettringite
C-S-H
CH
aggregate
Transition zone
HCP

4.3 Structure

Hardened Cement Paste (HCP)

Solids
- C-S-H
- CH
- Ettringite (AFt)
- Monosulfate hydrate (AFm)
- Residual unhydrated cement

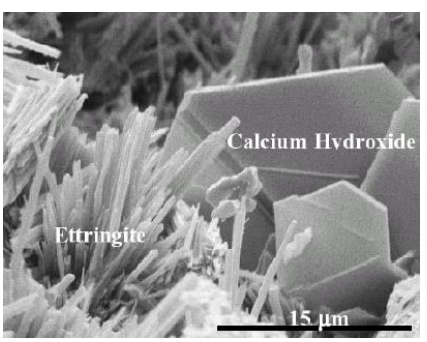
Voids
- Entrapped air (>1mm)
- Entrained air (75-500um)
- Capillary pores (macro →meso)
- Interlayer space (micropores)

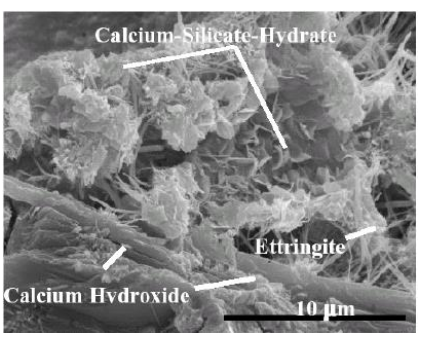
Water
- Capillary water
- Adsorbed water
- Interlayer water
- Chemically combined water

4.3 Structure

Hardened Cement Paste (HCP)

Solids





4.3 Structure

Hardened Cement Paste (HCP)

Solids

From Mehta and Monteiro 2006

Material	Volume Fraction of Solids (%)	Density (kg m)$^{-3}$	Surface Area ($m^2 g^{-1}$)	Morphology and Crystallinity
C—S—H	65	2000	400	Irregular foil, amorphous, microporous
CH	20	2250	~0.5	Thick hexagonal plates that cleave easily and are crystalline
$C_4A\bar{S}H_{12}$ (monosulfo-aliuminate)	10	1950	~2	Very thin irregular plates clustered in "rosettes"; fairly crystalline
Unhydrated residues	5	3150	0.1	Remnants of original cement grains

4.3 Structure

HCP

From Mehta and Monteiro 2006

Component	Strength	Deformations	Durability
C—S—H	Provides major cohesive force but is intrinsically weak because of micro-porosity. Dry concrete is stronger than wet (stronger van der Waals bonds).	Water loss from micro pores causes shrinkage (on drying) and creep (on loading) at room temperature.	Very insoluble. Water in micropores does not freeze and has low mobility (low permeability).
Calcium hydroxide	Contributes by reducing porosity. Cleavage may limit strength of high-strength pastes.	Is dimensionally stable. Will restrain C—S—H deformations.	Blocks capillary pores and hence lowers permeability. But slowly leached by water, which causes efflorescence and increases permeability. Carbonates, and is dissolved by acids.
$C_4A\bar{S}H_{12}$	Not significant. Reduces total porosity.	Minor effect.	Causes sulfate attack by reforming ettringite and causing expansion.
Unhydrated cement	Significant only in low-porosity pastes.	Will restrain C—S—H deformations.	Renewed hydration may cause autogeneous healing of internal microcracks.
Capillary pores	Total porosity is the major factor influencing strength.	Fine pores contribute to shrinkage and creep.	Porosity influences permeability and diffusivity. Large pores increase water flow through concrete.

4.3 Structure

Hardened Cement Paste (HCP)

Voids

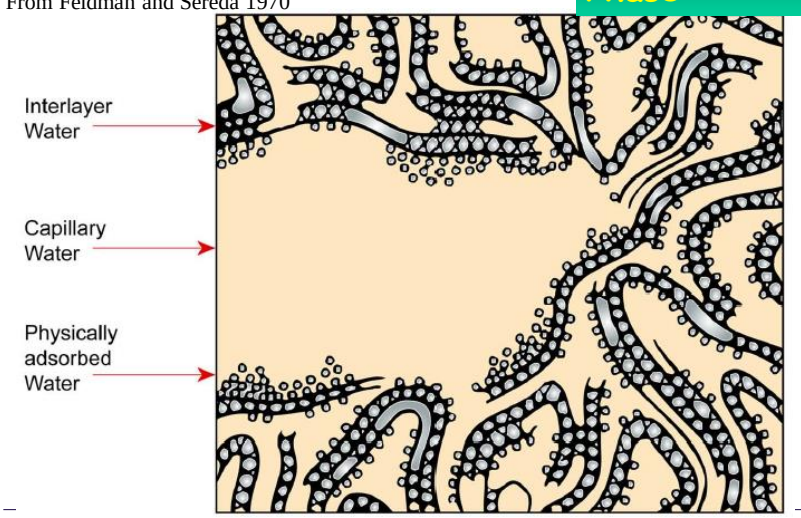
Voids are intrinsically part of the hydrated cement paste, due to the excess water used for economy and workability in the vast majority of mixtures.

- Entrapped air (>1mm)
- Entrained air (75-500um)
- Capillary pores (macro →meso)
- Interlayer space (micropores)

4.3 Structure

Hardened Cement Paste Phase

From Feldman and Sereda 1970



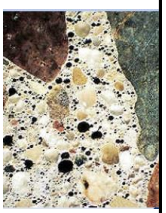
4.3 Structure

Hardened Cement Paste (HCP) **Voids**

C-S-H interlayer space

- 0.5-2.5 nm in size; too small to adversely affect strength or permeability

Capillary pores



- space not take up by the cement and hydration products;
- 2.5-50nm in size in well-hydrated concrete
- irregular in shape
- size and amount is related to w/c and degree of hydration

Entrained air

- spherical voids 75-500um in size; added for freeze/thaw resistance

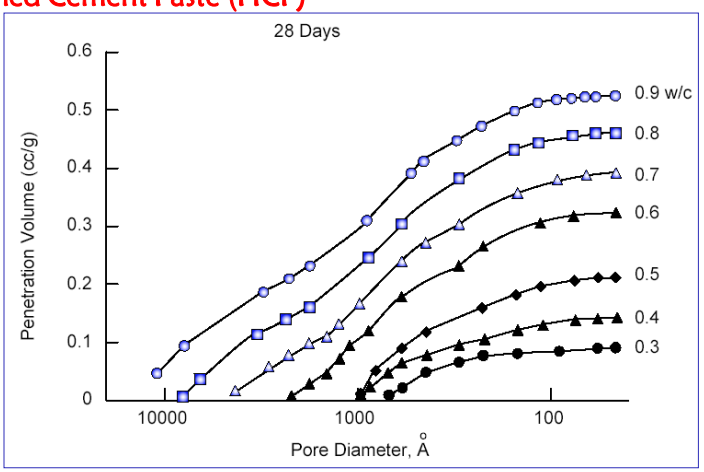
Entrapped air

- irregular in shape; can be large

4.3 Structure

Hardened Cement Paste (HCP)

Voids



From Mehta and Montairo 2006

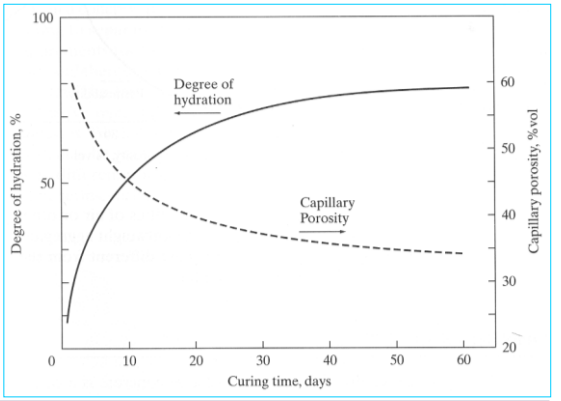
4.3 Structure

Structure and Performance

Voids

The presence of voids affects

- Strength
- Stress distribution (concentrations)
- permeability
- freeze/thaw resistance



4.3 Structure

Hardened Cement Paste (HCP)

Water

Hardened cement paste is never completely dry, it contains water of different natures: physical water (free, capillary and absorbed) and chemical water. One can drive them out of HCP by drying (for physical water) or heating (for chemical water.)

4.3 Structure

Hardened Cement Paste (HCP)

Water

- Ratio of mass of water to mass of cement in a mixture is the "water-to-cement ratio" or "w/c"
- When SCM's are used, this is the "water-to-cementitious materials ratio" or "w/cm"
- w/c or w/cm may range 0.2-0.8, but 0.4-0.6 is typical

Water is

- Introduced to the concrete during mixing
- Permeates the concrete during service
- Because the water in concrete contains ions, it is usually called "pore solution" and has a high pH

4.3 Structure

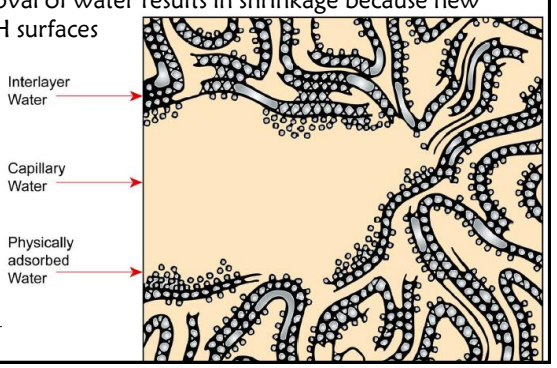
Hardened Cement Paste (HCP) Water

Capillary water

- water present in voids larger than 2.5nm
- In capillaries >50nm, water exists as "free water" because its removal does not cause volume change
- In capillaries 2.5-50nm, removal of water results in shrinkage because new bonds can form between C-S-H surfaces

Adsorbed water

- water physically adsorbed to the solid surfaces in C-S-H
- can be removed on drying to RH ~ 30%, resulting in shrinkage



4.3 Structure

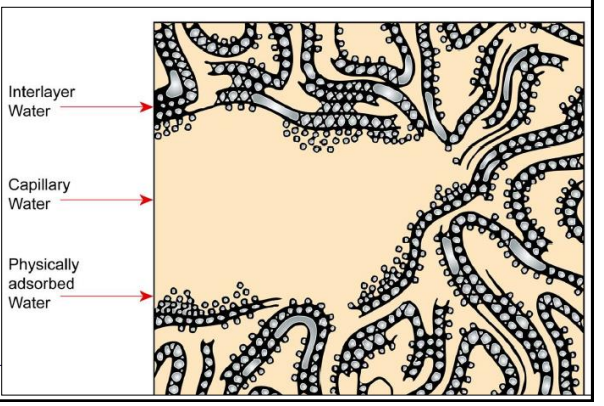
Hardened Cement Paste (HCP) Water

Interlayer water

- water associated with the C-S-H structure
- can be removed only on strong drying to RH ~ 11%, resulting in shrinkage

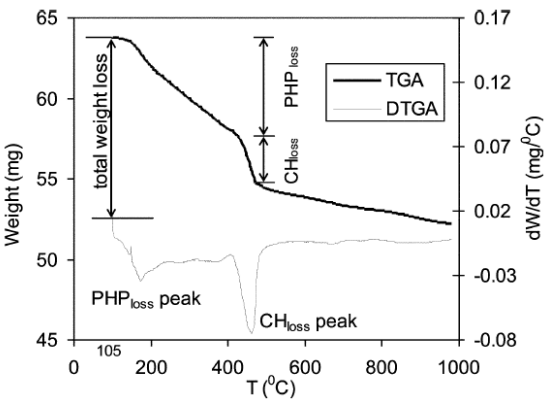
Chemically combined water

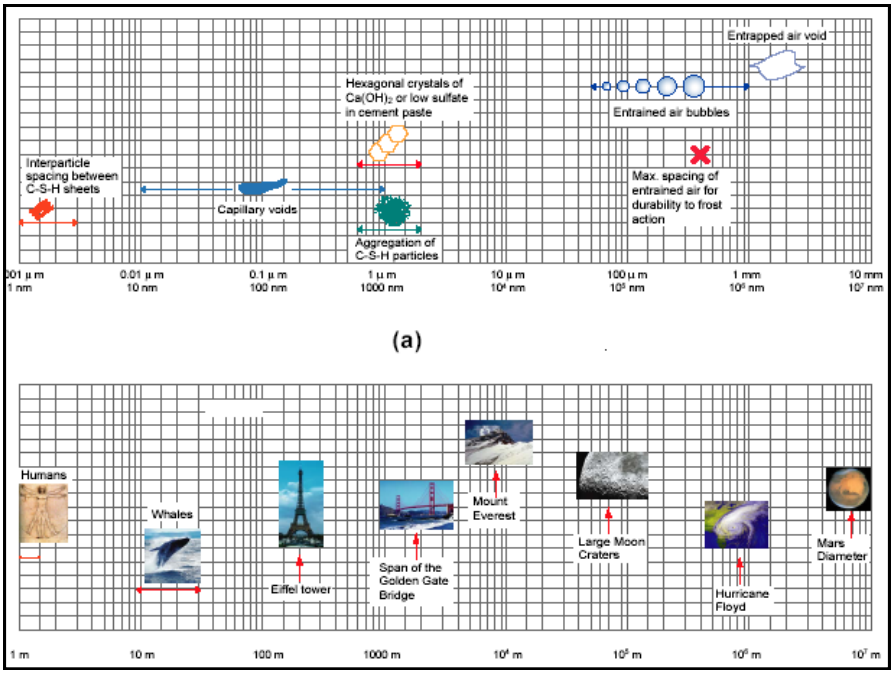
- water that is an integral part of various hydration products
- lost only on decomposition during heating



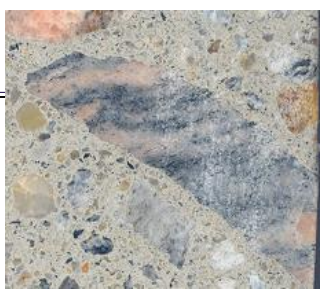
4.3 Structure

Hardened Cement Paste (HCP) Water





4.3 Structure



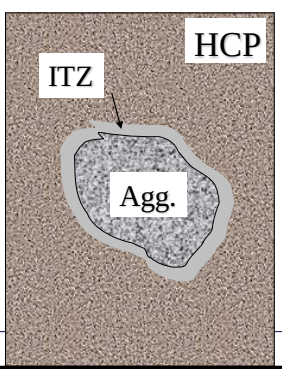
Interfacial Transition Zone, ITZ

The ITZ is the region 10-50um wide around coarse aggregate;

Characterized by:

- Higher local porosity
- Greater density of preexisting Microcracks
- Larger CH crystals that tend to be oriented and more prone to cleavage.

WEAK link in concrete!



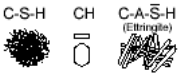
4.3 Structure

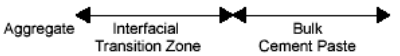
Interfacial Transition Zone, ITZ

ITZ: Weakest chain in concrete









4.3 Structure

Interfacial Transition Zone, ITZ

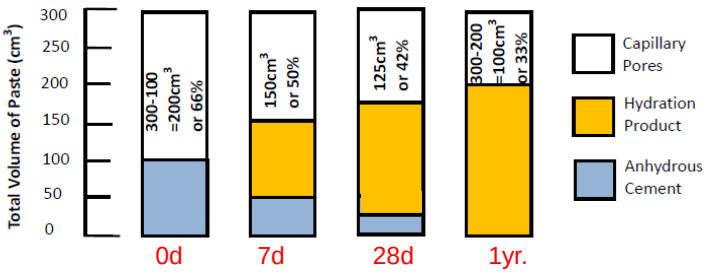
By tailoring the concrete mixture to reduce the influence of the ITZ. Then, strength, E, and impermeability are increased.

- ☐ lower w/c
- ☐ higher cement content
- ☐ use of SCMs
- ☐ lightweight aggregate
- ☐ extended moist curing



Calculating pore volume:

CASE A: 100 cm^3 of cement, constant w/c=0.63, varying degree of hydration as shown

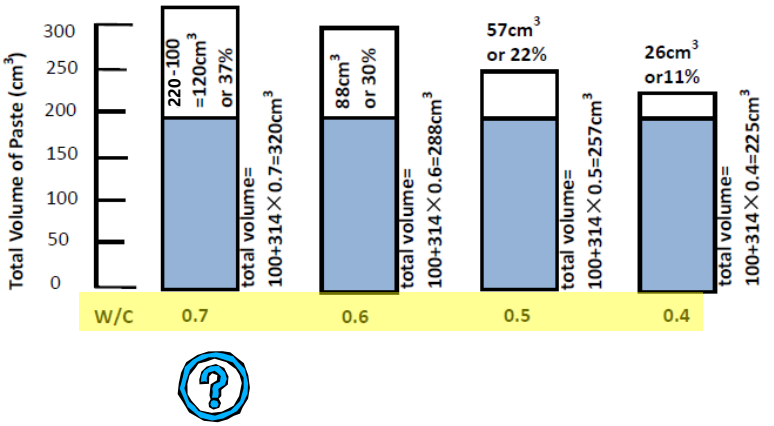


Days Hydrated	None		7d		28d		1yr.
Days Degree			50%		75%		100%

Assumption:
One volume of the cement + one volume of the water will produce two volume of products

Calculating pore volume:

CASE B: 100cm^3 of cement, 100% hydration, varying w/c as shown



4.3 Structure

Structure and Performance Voids - Strength

Inverse relationship between strength (f_c) and porosity (p)

$$f_c = a\,(1\text{-}p)^3$$

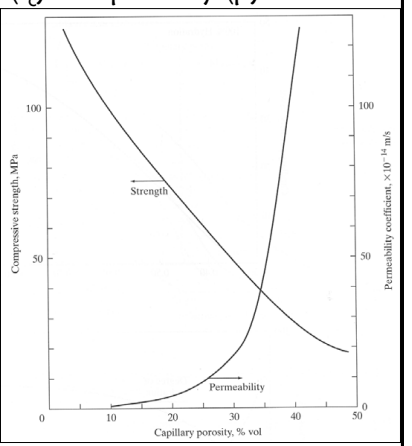
f_c: compressive strength of HCP

a: constant，~234 MPa

p: porosity

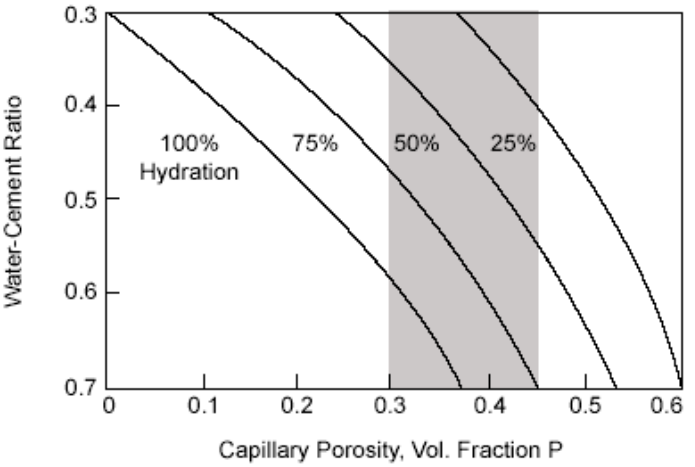
Porosity = volume of porosity/total volume

Powers and Brunauer, 1962



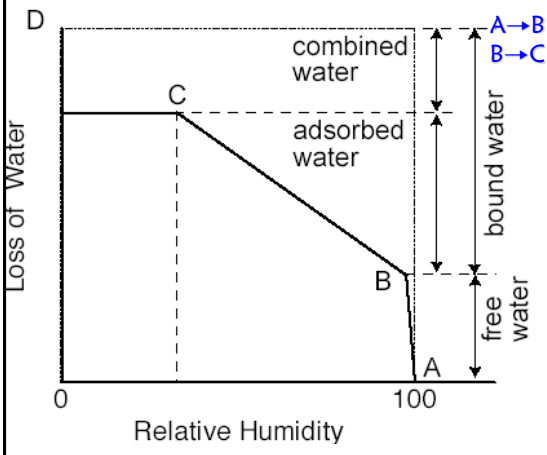
4.3 Structure

Structure and Performance Voids - Strength



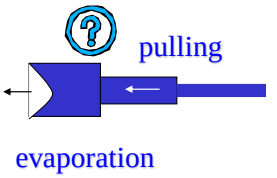
4.3 Structure

Structure and Performance



Structure - Deformation

A→B：free water evaporation
B→C：adsorption water evaporation



When a saturated cement paste is dried, the loss of water is not directly proportional to the drying shrinkage. Explain why.

Chapter Four：Concrete

1. Introduction
2. Compositions
3. Structure
4. **Strength and Failure**
5. Mix Proporationing
6. Early-age Properties
7. Volume Changes
8. Durability



4.4 Strength and Failure

Definition:

- The strength of a material is defined as the ability to resist stress without failure.
- Failure is identified with the appearance of cracks.
- However, it should be noted that unlike most structural materials, concrete contains fine cracks even before it is subjected to external stresses.



4.4 Strength and Failure

Universal strength-porosity relationship for solids :

$$S = S_0 \exp(-kp)$$

S： real strength
S_0： strength of solid with no voids
k： constant
p： porosity

Simplified as …

HCP

$$f_c = a\,(1\text{-}p)^3$$

Powers and Brunauer, 1962

4.4 Strength and Failure

Universal strength-porosity relationship

- Whereas in hardened cement paste or mortar the porosity can be related to strength, with concrete the situation is not simple.



- The presence of microcracks in the transition zone between the coarse aggregate and the cement paste matrix makes concrete too complex a material for prediction of strength by precise strength-porosity relations.

4.4 Strength and Failure

Failure modes in concrete



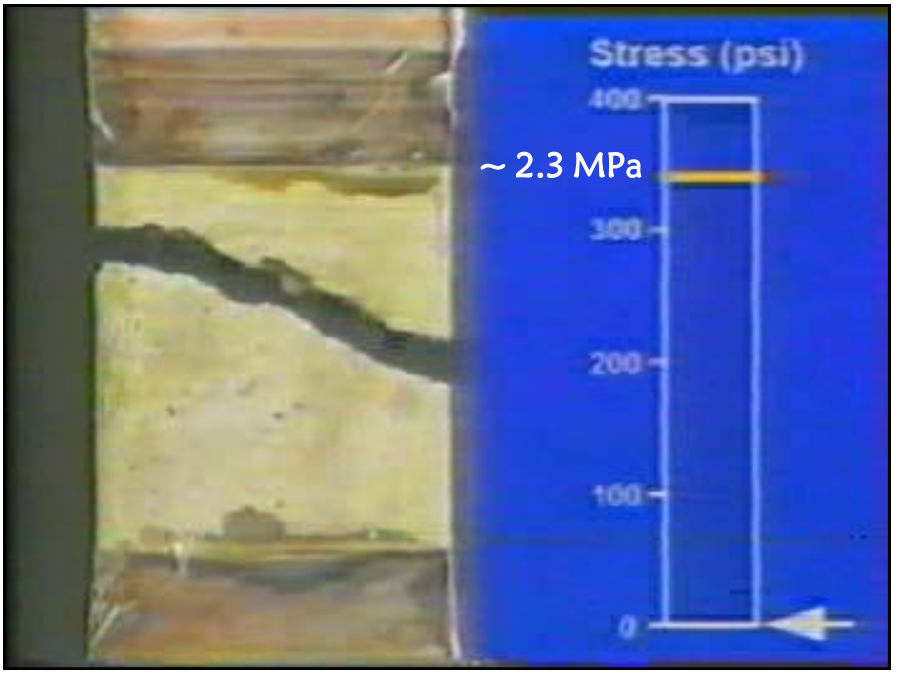
With a material such as concrete, which contains void spaces of various size and shape in the matrix and microcracks at the transition zone between the matrix and coarse aggregates, the failure modes under stress are very complex and vary with the type of stress.

4.4 Strength and Failure

Failure modes in concrete

- Under uniaxial tension, relatively less energy is need for the initiation and growth of cracks in the matrix. Rapid propagation and inter-linkage of the crack system, consisting or preexisting cracks at the transition zone and newly formed cracks in the matrix, account for the brittle failure.

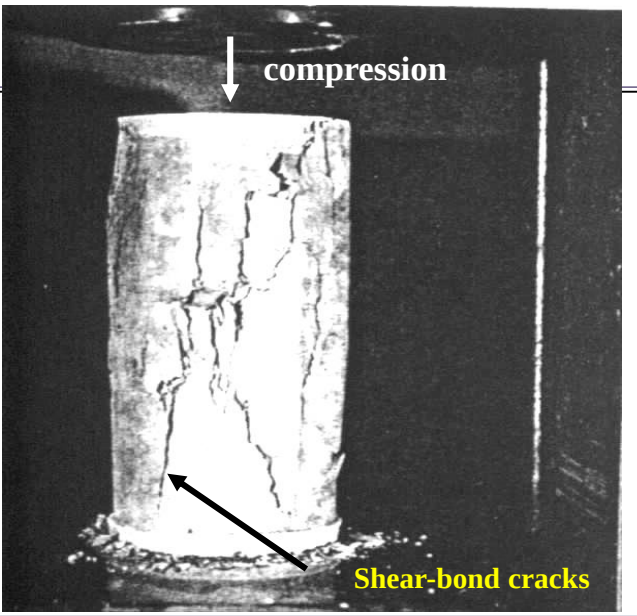
Brittle failure!



4.4 Strength and Failure

Failure modes in concrete

- In compression, the failure mode is less brittle because considerably more energy is needed to form and to extend cracks in the matrix.
- It is generally agreed that in a uniaxial compression test on medium-or low-strength concrete, no cracks are initiated in the matrix up to about 50 percent of the failure stress; at this stage a stable system of cracks, called shear-bond cracks, already exists in the vicinity of coarse aggregate.
- At higher stress levels, cracks are initiated within the matrix; their number and size increases progressively with increasing stress levels.
- The cracks in the matrix and the transition zone (shear-bond cracks) eventually join up, and generally a failure surface develops.

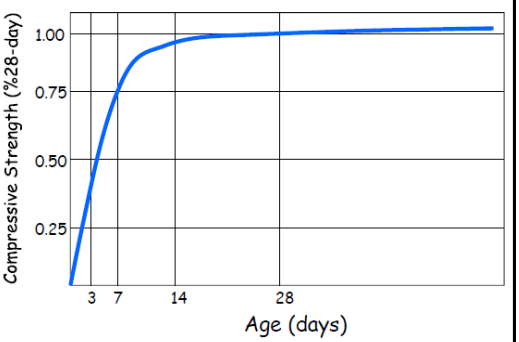

compression
Shear-bond cracks



4.4 Strength and Failure >> Compressive strength

Compressive Strength - is defined as the measured maximum resistance of a concrete or mortar specimen to an axial load at an age of 28-days.





4.4 Strength and Failure >> Compressive strength

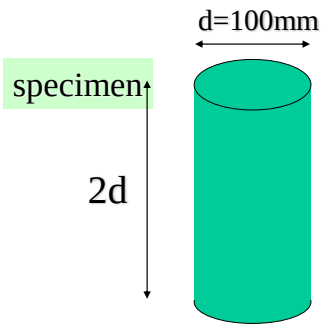
Compressive Strength



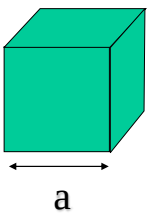


4.4 Strength and Failure >> compressive strength

Testing Method



China：a = 150 mm



cylinder	cube
（USA、France、Japan）	（UK、Germany、China）

4.4 Strength and Failure >> compressive strength

Factors affecting compressive strength

- Characteristics and proportions of materials
- Curing conditions
- Testing parameters

- During the first week to 10 days of curing it is important that the concrete not be permitted to freeze or dry out.



- In practical terms, about 90% of its strength is gained in the first 28 days.

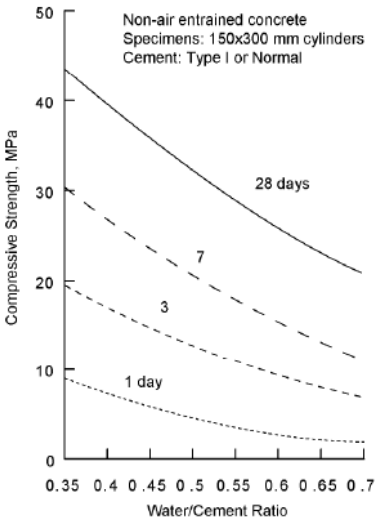
4.4 Strength and Failure >> compressive strength

Effect of Mix Proportions

Water-cement ratio rule, Abrams 1918

$$f_c = \frac{k_1}{k_2^{w/c}}$$

k_1,k_2 is constant



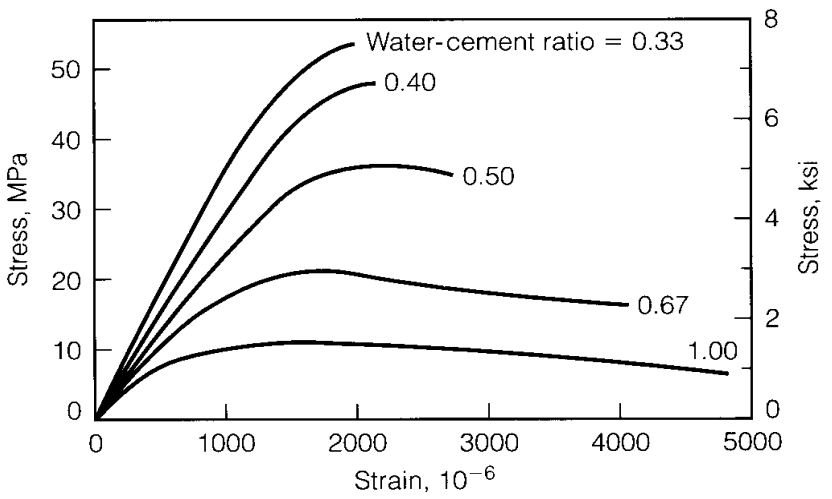


FIGURE 7.10 Typical stress-strain relations for compressive tests on 0.15 m by 0.30 m concrete cylinders with different water-cement values at an age of 28 days.

Summary

1. Porosity

Gel pore, Capillary pore, voids

Effect: strength

2. Water

Free water, adsorbed water, chemically bounded water

Effect: deformation

3. ITZ

Effect: strength

4. Strength

Compressive failure mode?

Term Project (Due: Dec. 14, 2018)

题材、字数、语言不限，打印后交上来 (No specific restriction on the theme, length, or language. Submission must be in printed form)

1. 正交异性钢桥面板的焊接工艺。
2. 寒冷地区混凝土技术与施工。
3. 新材料、新工艺、新结构研究进展。

Group Presentation (Dec. 14 and 21, 2018)



Questions?